```
mysql> show databases;
+---------------------------+
| Database                  |
+---------------------------+
| collage_db                |
| college_db                |
| information_schema        |
| library_management_system |
| mysql                     |
| performance_schema        |
| pharmadb                  |
| shop_stores               |
| sys                       |
+---------------------------+
9 rows in set (0.01 sec)

mysql> use library_management_system;
Database changed
```

MySQL 8.0 Command Line Cli

```
mysql> show databases;
+----------------------------+
| Database                   |
+----------------------------+
| collage_db                 |
| college_db                 |
| information_schema         |
| library_management_system  |
| mysql                      |
| performance_schema         |
| pharmadb                   |
| shop_stores                |
| sys                        |
+----------------------------+
9 rows in set (0.00 sec)

mysql> use library_management_system;
Database changed
mysql> create table Books(author_id int primary key, name varchar(100), country varchar(50));
Query OK, 0 rows affected (0.05 sec)

mysql> alter table books
    -> rename to Authors;
Query OK, 0 rows affected (0.04 sec)

mysql> desc Authors;
+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| author_id  | int          | NO   | PRI | NULL    |       |
| name       | varchar(100) | YES  |     | NULL    |       |
| country    | varchar(50)  | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
3 rows in set (0.01 sec)

mysql> create table Books(book_id int primary key, title varchar(150), author_id int , foreign key (author_id) references Authors(author_id), publish_year y
ear, available_copies int);
Query OK, 0 rows affected (0.08 sec)

mysql> desc books;
+-----------------+--------------+------+-----+---------+-------+
```

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```
mysql> desc books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| publish_year     | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> create table Members(member_id int primary key, name varchar(100), email varchar(100) unique , phone varchar(15));
Query OK, 0 rows affected (0.07 sec)

mysql> desc members;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| member_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| email     | varchar(100) | YES  | UNI | NULL    |       |
| phone     | varchar(15)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> create table Borrow_Records(record_id int primary key, member_id int , foreign key (member_id) references members(member_id));
Query OK, 0 rows affected (0.07 sec)

mysql> desc borrow_records;
+-----------+------+------+-----+---------+-------+
| Field     | Type | Null | Key | Default | Extra |
+-----------+------+------+-----+---------+-------+
| record_id | int  | NO   | PRI | NULL    |       |
| member_id | int  | YES  | MUL | NULL    |       |
+-----------+------+------+-----+---------+-------+
2 rows in set (0.01 sec)

mysql> alter table borrow_records
    -> add column book_id int,
    -> foreign key (book_id) references books(book_id),
```

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MySQL 8.0 Command Line Cli

```
    -> add column book_id int,
    -> foreign key (book_id) references books(book_id),
    -> \c
mysql> drop table Borrow_records;
Query OK, 0 rows affected (0.03 sec)

mysql> create table borrow_records(record_id int primary key, member_id int , foreign key (member_id) references members(member_id), book_id int, foreign ke
y (book_id) references books(book_id), borrow_date date, return_date date);
Query OK, 0 rows affected (0.09 sec)

mysql> desc borrow_records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  | MUL | NULL    |       |
| book_id     | int  | YES  | MUL | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> alter table books
    -> add column genre varchar(50);
Query OK, 0 rows affected (0.16 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| publish_year     | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
| genre            | varchar(50)  | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> show tables;
```

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MySQL 8.0 Command Line Cli

```
mysql> show tables;
+-----------------------------------+
| Tables_in_library_management_system |
+-----------------------------------+
| authors                           |
| books                             |
| borrow_records                    |
| members                           |
+-----------------------------------+
4 rows in set (0.00 sec)

mysql> insert into authors(author_id,name,country)
    -> values
    -> (1,'chetan bhagat','india'),
    -> (2,'apj abdul kalam',india),
    -> (3,'rk narayan','india');
ERROR 1054 (42S22): Unknown column 'india' in 'field list'
mysql> insert into authors(author_id,name,country)
    -> values
    -> (1,'chetan bhagat','india'),
    -> (2,'apj abdul kalam','india'),
    -> (3,'rk narayan','india');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from authors;
+-----------+-----------------+---------+
| author_id | name            | country |
+-----------+-----------------+---------+
|         1 | chetan bhagat   | india   |
|         2 | apj abdul kalam | india   |
|         3 | rk narayan      | india   |
+-----------+-----------------+---------+
3 rows in set (0.00 sec)

mysql> insert into books(book_id,title,author_id,published_year,available_copies)
    -> values
    -> (1,'Wings of fire',2,1999,100),
    -> (2,'Malgudi Days',3,1943,150),
    -> (3,'One Night @ the Call Center',1,2005,500),
```

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MySQL 8.0 Command Line Cli

```
    -> (5,'shyam chi aai',10,1935,120);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`library_management_system`.`books`, CONSTRAINT `books_ibfk_1` FOREIGN
 KEY (`author_id`) REFERENCES `authors` (`author_id`))
mysql> insert into books(book_id,title,author_id,publish_year,available_copies)
    -> values
    ->  (1,'Wings of fire',2,1999,100),
    -> (2,'Malgudi Days',3,1943,150),
    -> (3,'One Night @ the Call Center',1,2005,500),
    -> (4,'swami and friends',3,1935,200),
    -> (5,'five point someone',1,2004,120);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from books;
+---------+---------------------------+-----------+--------------+------------------+-------+
| book_id | title                     | author_id | publish_year | available_copies | genre |
+---------+---------------------------+-----------+--------------+------------------+-------+
|       1 | Wings of fire             |         2 |         1999 |              100 | NULL  |
|       2 | Malgudi Days              |         3 |         1943 |              150 | NULL  |
|       3 | One Night @ the Call Center |         1 |         2005 |              500 | NULL  |
|       4 | swami and friends         |         3 |         1935 |              200 | NULL  |
|       5 | five point someone        |         1 |         2004 |              120 | NULL  |
+---------+---------------------------+-----------+--------------+------------------+-------+
5 rows in set (0.00 sec)

mysql> insert into members(member_id,name,email,phone)
    -> values
    -> (2003,'Nayan Chaudhari','nayanchaudhari@gmail.com',7559384586),
    -> (1944,'Niraj Dandge','nirajdandge@gmail.com',8071905245),
    -> (5202,'Aman Khandare','amankhandare@gmail.com',9052445563);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from members;
+-----------+----------------+-------------------------+------------+
| member_id | name           | email                   | phone      |
+-----------+----------------+-------------------------+------------+
|      1944 | Niraj Dandge   | nirajdandge@gmail.com   | 8071905245 |
|      2003 | Nayan Chaudhari | nayanchaudhari@gmail.com | 7559384586 |
|      5202 | Aman Khandare  | amankhandare@gmail.com  | 9052445563 |
+-----------+----------------+-------------------------+------------+
```

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MySQL 8.0 Command Line Cli

```
|      5202 | Aman Khandare   | amankhandare@gmail.com   | 9052445563 |
+-----------+-----------------+--------------------------+------------+
3 rows in set (0.00 sec)

mysql> show tabels;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't
abels' at line 1
mysql> show tables;
+-----------------------------------+
| Tables_in_library_management_system |
+-----------------------------------+
| authors                           |
| books                             |
| borrow_records                    |
| members                           |
+-----------------------------------+
4 rows in set (0.00 sec)

mysql> insert into borrow_records(record_id,member_id,book_id,borrow_date,return_date)
    -> values
    -> (102,5202,3,'2026-10-10','2026-11-10'),
    -> (103,2003,1,'2026-10-5','2027-1-20'),
    -> (104,1944,1,'2026-10-6','2027-2-5'),
    -> (105,5202,5,'2026-9-10','2027-12-3');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|       102 |      5202 |       3 | 2026-10-10  | 2026-11-10  |
|       103 |      2003 |       1 | 2026-10-05  | 2027-01-20  |
|       104 |      1944 |       1 | 2026-10-06  | 2027-02-05  |
|       105 |      5202 |       5 | 2026-09-10  | 2027-12-03  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql> select * from books
    -> where publish_year>2015;
Empty set (0.00 sec)
```

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MySQL 8.0 Command Line Cli

```
mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|       102 |      5202 |       3 | 2026-10-10  | 2026-11-10  |
|       103 |      2003 |       1 | 2026-10-05  | 2027-01-20  |
|       104 |      1944 |       1 | 2026-10-06  | 2027-02-05  |
|       105 |      5202 |       5 | 2026-09-10  | 2027-12-03  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql> select * from books
    -> where publish_year>2015;
Empty set (0.00 sec)

mysql> select * from books;
+---------+-----------------------------+-----------+--------------+------------------+-------+
| book_id | title                       | author_id | publish_year | available_copies | genre |
+---------+-----------------------------+-----------+--------------+------------------+-------+
|       1 | Wings of fire               |         2 |         1999 |              100 | NULL  |
|       2 | Malgudi Days                |         3 |         1943 |              150 | NULL  |
|       3 | One Night @ the Call Center |         1 |         2005 |              500 | NULL  |
|       4 | swami and friends           |         3 |         1935 |              200 | NULL  |
|       5 | five point someone          |         1 |         2004 |              120 | NULL  |
+---------+-----------------------------+-----------+--------------+------------------+-------+
5 rows in set (0.00 sec)

mysql> select * from books
    -> where publish_year>2015;
Empty set (0.00 sec)

mysql>
```

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MySQL 8.0 Command Line Cli

```
mysql> show databases;
+----------------------------+
| Database                   |
+----------------------------+
| collage_db                 |
| college_db                 |
| information_schema         |
| library_management_system  |
| mysql                      |
| performance_schema         |
| pharmadb                   |
| shop_stores                |
| sys                        |
+----------------------------+
9 rows in set (0.06 sec)

mysql> use library_management_system;
Database changed
mysql> show tables;
+-----------------------------------+
| Tables_in_library_management_system |
+-----------------------------------+
| authors                           |
| books                             |
| borrow_records                    |
| members                           |
+-----------------------------------+
4 rows in set (0.05 sec)

mysql> select * from authors;
+-----------+-----------------+---------+
| author_id | name            | country |
+-----------+-----------------+---------+
|         1 | chetan bhagat   | india   |
|         2 | apj abdul kalam | india   |
|         3 | rk narayan      | india   |
+-----------+-----------------+---------+
3 rows in set (0.05 sec)

mysql> select * from books;
+---------+---------------------------+-----------+--------------+------------------+-------+
```

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MySQL 8.0 Command Line Cli

```
3 rows in set (0.05 sec)

mysql> select * from books;
+---------+-----------------------------+-----------+--------------+------------------+-------+
| book_id | title                       | author_id | publish_year | available_copies | genre |
+---------+-----------------------------+-----------+--------------+------------------+-------+
|       1 | Wings of fire               |         2 |         1999 |              100 | NULL  |
|       2 | Malgudi Days                |         3 |         1943 |              150 | NULL  |
|       3 | One Night @ the Call Center |         1 |         2005 |              500 | NULL  |
|       4 | swami and friends           |         3 |         1935 |              200 | NULL  |
|       5 | five point someone          |         1 |         2004 |              120 | NULL  |
+---------+-----------------------------+-----------+--------------+------------------+-------+
5 rows in set (0.03 sec)

mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|       102 |      5202 |       3 | 2026-10-10  | 2026-11-10  |
|       103 |      2003 |       1 | 2026-10-05  | 2027-01-20  |
|       104 |      1944 |       1 | 2026-10-06  | 2027-02-05  |
|       105 |      5202 |       5 | 2026-09-10  | 2027-12-03  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.03 sec)

mysql> select * from members;
+-----------+-----------------+--------------------------+-------------+
| member_id | name            | email                    | phone       |
+-----------+-----------------+--------------------------+-------------+
|      1944 | Niraj Dandge    | nirajdandge@gmail.com    | 8071905245  |
|      2003 | Nayan Chaudhari | nayanchaudhari@gmail.com | 7559384586  |
|      5202 | Aman Khandare   | amankhandare@gmail.com   | 9052445563  |
+-----------+-----------------+--------------------------+-------------+
3 rows in set (0.03 sec)

mysql> select * from
    -> borrow_records , members
    -> where member_id=2003;
ERROR 1052 (23000): Column 'member_id' in where clause is ambiguous
mysql> select b.member_id,b.book_id,m.name from
    -> borrow_records as b , members as m
```

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MySQL 8.0 Command Line Cli

```
mysql> select * from members;
+-----------+-----------------+--------------------------+------------+
| member_id | name            | email                    | phone      |
+-----------+-----------------+--------------------------+------------+
|      1944 | Niraj Dandge    | nirajdandge@gmail.com    | 8071905245 |
|      2003 | Nayan Chaudhari | nayanchaudhari@gmail.com | 7559384586 |
|      5202 | Aman Khandare   | amankhandare@gmail.com   | 9052445563 |
+-----------+-----------------+--------------------------+------------+
3 rows in set (0.03 sec)

mysql> select * from
    -> borrow_records , members
    -> where member_id=2003;
ERROR 1052 (23000): Column 'member_id' in where clause is ambiguous
mysql> select b.member_id,b.book_id,m.name from
    -> borrow_records as b , members as m
    -> where b.member_id=m.member_id;
+-----------+---------+-----------------+
| member_id | book_id | name            |
+-----------+---------+-----------------+
|      1944 |       1 | Niraj Dandge    |
|      2003 |       1 | Nayan Chaudhari |
|      5202 |       3 | Aman Khandare   |
|      5202 |       5 | Aman Khandare   |
+-----------+---------+-----------------+
4 rows in set (0.01 sec)

mysql> select b.member_id,b.book_id,m.name ,bk.title from
    -> borrow_records as b , members as m , books as bk
    -> where b.member_id=m.member_id and b.book_id=bk.book_id;
+-----------+---------+-----------------+-----------------------------+
| member_id | book_id | name            | title                       |
+-----------+---------+-----------------+-----------------------------+
|      1944 |       1 | Niraj Dandge    | Wings of fire               |
|      2003 |       1 | Nayan Chaudhari | Wings of fire               |
|      5202 |       3 | Aman Khandare   | One Night @ the Call Center |
|      5202 |       5 | Aman Khandare   | five point someone          |
+-----------+---------+-----------------+-----------------------------+
4 rows in set (0.00 sec)

mysql>
```

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```
|       103 |      2003 |       1 | 2026-10-05  | 2027-01-26  |
|       104 |      1944 |       1 | 2026-10-06  | 2027-02-05  |
|       105 |      5202 |       5 | 2026-09-10  | 2027-12-03  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql> select * from members;
+-----------+-----------------+---------------------------+-------------+
| member_id | name            | email                     | phone       |
+-----------+-----------------+---------------------------+-------------+
|      1944 | Niraj Dandge    | nirajdandge@gmail.com     | 8071905245  |
|      2003 | Nayan Chaudhari | nayanchaudhari@gnail.com  | 7559384586  |
|      5202 | Aman Khandare   | amankhandare@gmail.com    | 9052445563  |
+-----------+-----------------+---------------------------+-------------+
3 rows in set (0.00 sec)

mysql>
mysql> update books
    -> where \c
mysql> update books
    -> set title='i don't love you anymore'
    '> where book_id=3;
    '>
    '> \c
    '> \c
    '> ^C
mysql> updage books
    -> set title='i do not love you anymore'
    -> where book_id=3;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'updage books
set title='i do not love you anymore'
where book_id=3' at line 1
mysql> update books
    -> set title='i do not love you anymore'
    -> where book_id=3;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from books;
+---------+---------------------------+-----------+--------------+------------------+-------+
| book_id | title                     | author_id | publish_year | available_copies | genre |
+---------+---------------------------+-----------+--------------+------------------+-------+
|       1 | Wings of fire             |         2 |         1999 |              100 | NULL  |
|       2 | Malgudi Days              |         3 |         1943 |              150 | NULL  |
|       3 | i do not love you anymore |         1 |         2005 |              500 | NULL  |
|       4 | swami and friends         |         3 |         1935 |              200 | NULL  |
|       5 | five point someone        |         1 |         2004 |              120 | NULL  |
+---------+---------------------------+-----------+--------------+------------------+-------+
5 rows in set (0.00 sec)

mysql> select title , name
    -> from books as b , authors as a
    -> where b.author_id=a.author_id;
+---------------------------+-----------------+
| title                     | name            |
+---------------------------+-----------------+
| i do not love you anymore | chetan bhagat   |
| five point someone        | chetan bhagat   |
| Wings of fire             | apj abdul kalam |
| Malgudi Days              | rk narayan      |
| swami and friends         | rk narayan      |
+---------------------------+-----------------+
5 rows in set (0.01 sec)
```

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MySQL 8.0 Command Line Cli

```
| borrow_records                     |
| members                            |
+------------------------------------+
4 rows in set (0.00 sec)

mysql> select * from authors;
+-----------+-----------------+---------+
| author_id | name            | country |
+-----------+-----------------+---------+
|         1 | chetan bhagat   | india   |
|         2 | apj abdul kalam | india   |
|         3 | rk narayan      | india   |
+-----------+-----------------+---------+
3 rows in set (0.00 sec)

mysql> select * from books;
+---------+---------------------------+-----------+--------------+------------------+-------+
| book_id | title                     | author_id | publish_year | available_copies | genre |
+---------+---------------------------+-----------+--------------+------------------+-------+
|       1 | Wings of fire             |         2 |         1999 |              100 | NULL  |
|       2 | Malgudi Days              |         3 |         1943 |              150 | NULL  |
|       3 | i do not love you anymore |         1 |         2005 |              500 | NULL  |
|       4 | swami and friends         |         3 |         1935 |              200 | NULL  |
|       5 | five point someone        |         1 |         2004 |              120 | NULL  |
+---------+---------------------------+-----------+--------------+------------------+-------+
5 rows in set (0.00 sec)

mysql> select * from borrow_records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|       102 |      5202 |       3 | 2026-10-10  | 2026-11-10  |
|       103 |      2003 |       1 | 2026-10-05  | 2027-01-20  |
|       104 |      1946 |       1 | 2026-10-06  | 2027-02-05  |
|       105 |      5202 |       5 | 2026-09-10  | 2027-12-03  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)

mysql> select * from members;
+-----------+----------------+---------------------------+------------+
| member_id | name           | email                     | phone      |
+-----------+----------------+---------------------------+------------+
|      1946 | Niraj Dandge   | nirajdandge@gmail.com     | 8071905245 |
|      2003 | Nayan Chaudhari | nayanchaudhari@gmail.com | 7559384586 |
|      5202 | Aman Khandare  | amankhandare@gmail.com    | 9057445563 |
+-----------+----------------+---------------------------+------------+
3 rows in set (0.00 sec)

mysql> delete * from books
    -> where publish_year<2000;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '* from books
where publish_year<2000' at line 1
mysql> delete * from books
    -> where publish_year = 1999;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '* from books
where publish_year = 1999' at line 1
mysql> delete from books
    -> where publish_year<2000;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`library_management_system`.`borrow_records`, CONSTRAINT `borrow_records_ibfk_2` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`))
mysql>
```

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